

MELSS OIS

Optical Inspection System

User Manual

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NOTE

About this manual: This document describes the MELSS Optical Inspection System as implemented in version 7.3. The application has six main tabs: RUN (live inspection), TRAIN (labelling and training), FILTERS (image pipeline design), AOI (offline batch inspection), INFER (standalone detection), and HISTORY (inspection records).

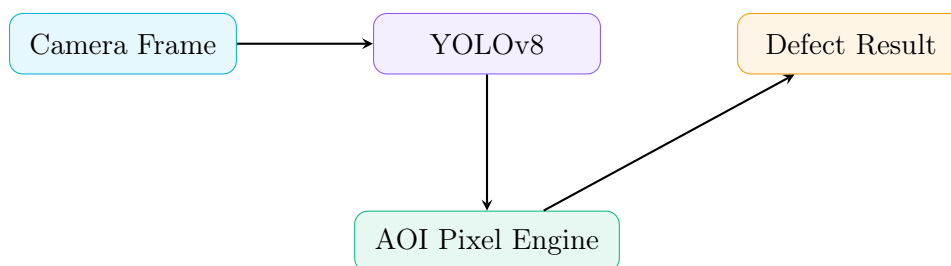
Note: The current version does not include an ROI (Region of Interest) editor. All inspection operates on the full camera frame.

1 System Architecture

NOTE

Why two engines? MELSS OIS combines deep-learning object detection with classical computer vision.

MELSS OIS uses a two-layer inspection strategy on every frame.



Layer 1 — YOLOv8 (AI): Detects components and outputs bounding boxes with class labels (Resistor, Capacitor, IC, etc.) and confidence scores.

Layer 2 — AOI Pixel Engine (CV): For each golden slot, computes variance (empty?), SSIM (structural similarity), NCC (patch matching), and rotational NCC (misalignment). Classifies as OK, MISSING, WRONG COMPONENT, WRONG POLARITY, or MISALIGNED.

2 Getting Started

2.1 Launching the Application

```
python optical_inspection_system.py
```

First launch creates a per-OS data directory:

- **Windows:** C:\Users\<user>\AppData\Local\MELSS\OIS
- **macOS:** ~/Library/Application Support/OIS
- **Linux:** ~/.local/share/OIS

2.2 Navigation Sidebar

Icon	Tab	Shortcut	Purpose
	RUN	F1	Live camera inspection
	TRAIN	F2	Label images, train the model
	FILTERS	F3	Design image filter pipeline
	AOI	F4	Offline batch inspection
	INFER	F5	Standalone YOLO detection
	HISTORY	F6	View past inspection records
	SETTINGS	—	Project-wide settings

2.3 Projects

A project folder contains:

- `golden_boards/` — labelled training images
- `labels/` — YOLO format label files
- `models/` — trained model weights (.pt)
- `config.json` — project configuration
- `golden_master.json` — component list for AOI comparison

3 RUN Tab: Live Camera Inspection

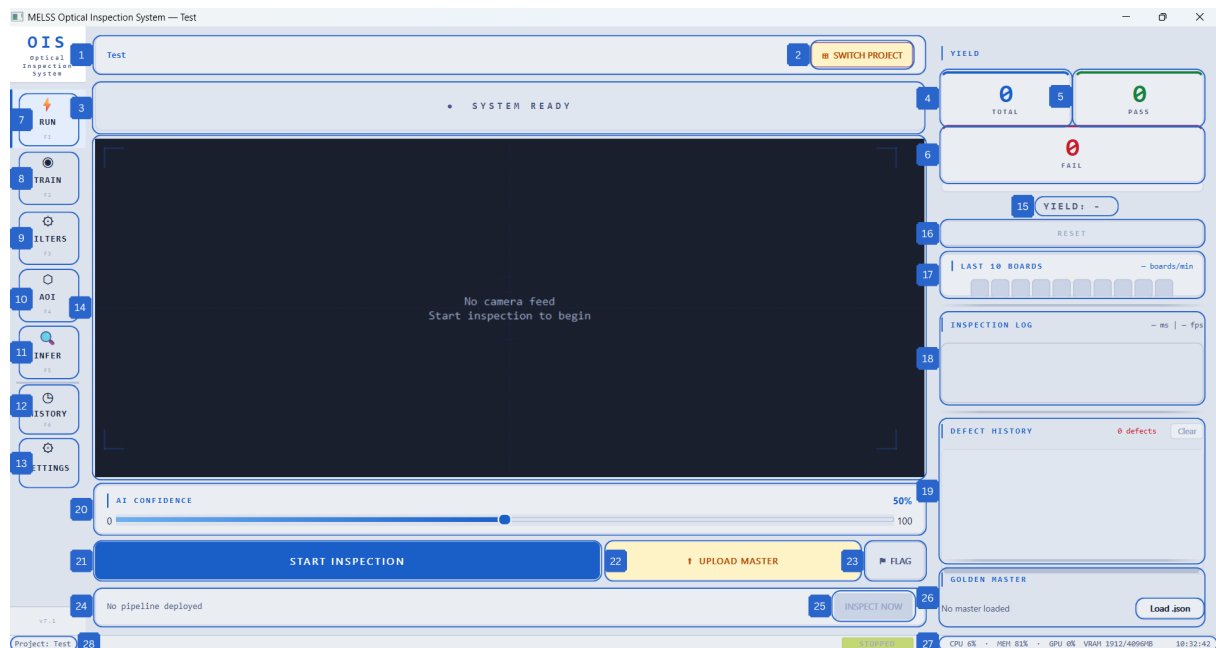


Figure 1: RUN Tab — Live Camera Inspection

NOTE

Purpose: Connect a camera, stream live PCB images, and automatically flag defects against a stored golden reference in real time.

3.1 Element Reference

Element Description	
1	Logo — Static brand mark. Version number in bottom-left of sidebar.
2	SWITCH PROJECT — Opens Project Manager. Create, open, or delete projects.
3	RUN Tab (active) — Currently selected tab.
4	Corner Brackets — Visual alignment guides for board positioning.
5	TOTAL / PASS Counters — Session counters (boards inspected / boards with zero defects).
6	Corner Bracket (bottom-right) — Fourth alignment guide.
7 12	Tab Icons — Navigation: RUN (F1), TRAIN (F2), FILTERS (F3), AOI (F4), INFER (F5), HISTORY (F6).
13	SETTINGS — Opens Settings panel (model path, confidence, SAHI, calibration).

	Element Description
14	AOI Icon — Duplicate visible at scroll position.
15	YIELD Display — Pass rate with progress bar. Green >50%, Amber 20-50%, Red <20%.
16	RESET — Resets session counters (does not delete history).
17	LAST 10 BOARDS — Coloured squares showing recent results (green=PASS, red=FAIL).
18	INSPECTION LOG — Scrolling log with timestamps, detections, AOI signals, errors.
19	FAIL Counter — Boards with at least one defect.
20	AI CONFIDENCE Slider — YOLO threshold (0-100%), recommended 40-60%.
21	START INSPECTION — Opens camera, starts streaming. Toggles to STOP.
22	UPLOAD MASTER — Dialog with three options: picture from disk, JSON file, or live camera frame.
23	FLAG — Saves current frame to <code>golden_boards/</code> for training.
24	Pipeline Status — Shows deployed filter pipeline status.
25	INSPECT NOW — Single manual inspection on current frame.
26	GOLDEN MASTER — Shows loaded golden master. Load .json to select reference.
27	Status Bar — Shows tab/project name, camera status, CPU/MEM/GPU metrics, clock.
28	Project Name — Current project name in status bar.

3.2 Workflow

1. **Start camera:** Click **START INSPECTION** [21](#)
2. **Load golden reference:** Click **UPLOAD MASTER** [22](#) and choose:
 - "Take Latest Frame" — capture from live camera (camera must be running)
 - "Choose Picture" — load from disk
 - "Upload JSON" — load existing `golden_master.json`
3. **Adjust confidence:** Set **AI CONFIDENCE** [20](#) (start at 50%)
4. **Inspect boards:** Present boards to camera. Watch YIELD display [15](#)
5. **Flag issues:** Click **FLAG** [23](#) to save problematic boards for training
6. **Manual check:** Click **INSPECT NOW** [25](#) for single board verification
7. **Stop:** Click STOP (same button) when done

8. **Reset batch:** Click **RESET** [16](#) for new batch statistics

TIP

Production tip: Keep AI CONFIDENCE between 40-60%. Lower if you see `var_rescue` messages (YOLO missing components). Higher if false positives are too frequent.

4 TRAIN Tab: Labelling and Training

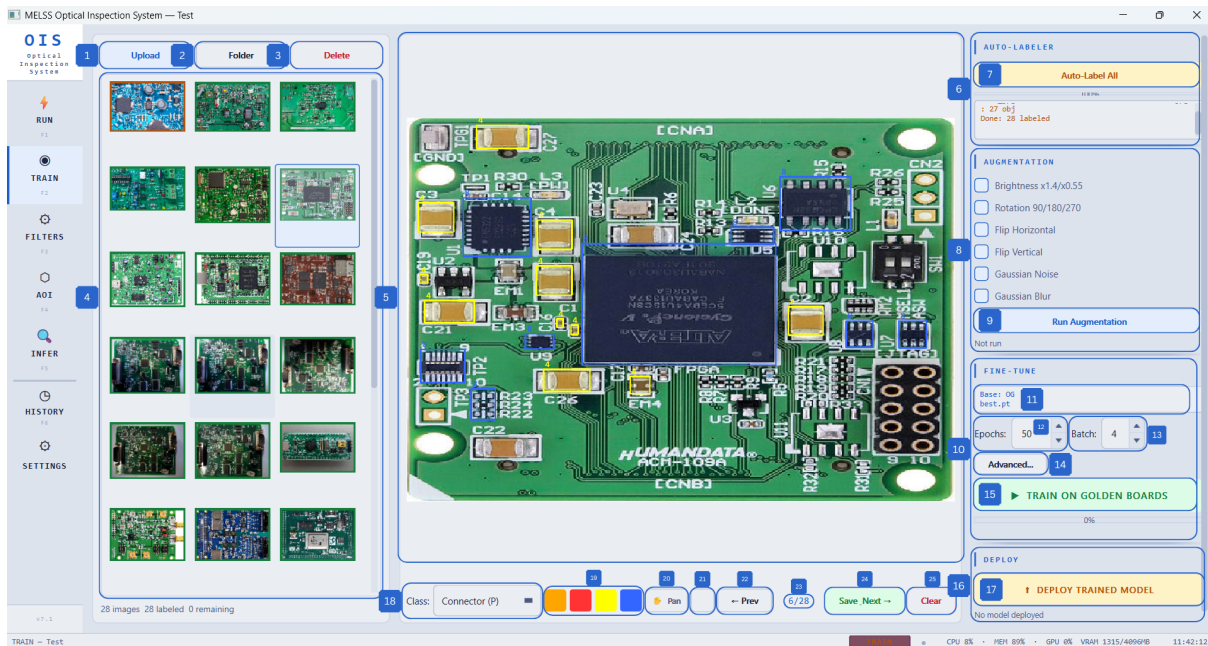


Figure 2: TRAIN Tab — Labelling and Training

NOTE

Purpose: Upload PCB images, draw bounding-box labels, configure augmentation, and fine-tune the YOLOv8 model.

4.1 Element Reference

Element Description	
1	Upload / Folder / Delete — Add images, open folder in OS, delete current image and label.
2	Upload Button — Primary way to add images. Supports drag-and-drop.
3	Folder Button — Opens <code>golden_boards/</code> in file system.
4	Thumbnail Grid — Border colours: green=labelled, red=unlabelled, cyan=selected.

Element Description	
5	Canvas Area — Draw boxes: select class, click-drag-release. Edit: click to select, drag to move, Delete to remove.
6	Right Panel — Four sections: AUTO-LABELER, AUGMENTATION, FINE-TUNE, DEPLOY.
7	Auto-Label All — Runs YOLO on all unlabelled images. Background thread.
8	Augmentation Checkboxes — Brightness, Rotation, Flip H, Flip V, Noise, Blur.
9	Run Augmentation — Generates augmented copies using checked operations.
10	FINE-TUNE Section — Training: Base model, Epochs, Batch size.
11	Base Model — Shows warm-start .pt file. Automatically chains previous training.
12	Epochs — Training passes (50 quick, 120 production).
13	Batch — Images per gradient update (4-8 for 4GB GPU).
14	Advanced — Full YOLOv8 hyperparameter dialog.
15	TRAIN — Starts fine-tune using project dataset.
16	DEPLOY Section — Deploy trained model to active configuration.
17	DEPLOY — Copies <code>best.pt</code> to <code>config.json</code> for all tabs.
18	Class Selector — 25 component classes for next box.
19	Quick-Class — P (orange), R (red), C (yellow), IC (blue).
20	Pan Toggle — Toggles pan mode. Hold Space for temporary pan.
21	Prev — Previous image.
22	Image Counter — Current position (<code>N/Total</code>).
23	Save & Next — Saves label and advances.
24	Clear — Removes all boxes from current image.
25	Summary — Progress: <code>N images</code> <code>N labeled</code> <code>N remaining</code> .

4.2 Workflow

1. **Add images:** Click **Upload** [2](#) or drag files into the thumbnail grid
2. **Auto-label:** Click **Auto-Label All** [7](#) to generate initial labels
3. **Review labels:** Click each thumbnail [4](#), check boxes on canvas [5](#)
 - Green border = labelled, Red border = needs labels
 - Right-click a box to delete, or drag to reposition

4. **Select class:** Use dropdown [18](#) or quick buttons [19](#) (P, R, C, IC)
5. **Save progress:** Click **Save & Next** [23](#) after each image
6. **Augment (optional):** Check augmentation options [8](#), click **Run Augmentation** [9](#)
7. **Configure training:** Set **Epochs** [12](#) (50 for quick test, 120 for production)
8. **Train:** Click **TRAIN** [15](#) and wait for completion (watch progress bar)
9. **Deploy:** Click **DEPLOY** [17](#) to use the new model in RUN tab

TIP

Training tip: Minimum 20 labelled images before training. Run Auto-Label first, then spend 15 minutes correcting boxes. Quality labels = quality detections.

5 FILTERS Tab: Image Pipeline Designer

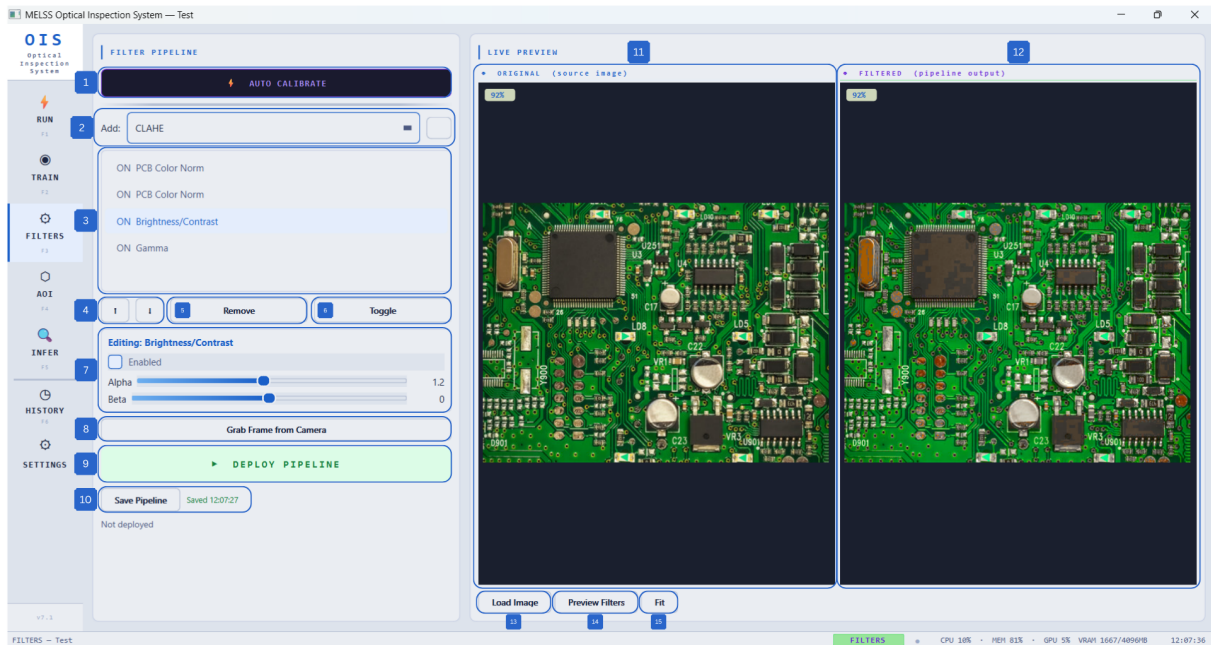


Figure 3: FILTERS Tab — Image Pipeline Designer

NOTE

Purpose: Build an image pre-processing pipeline that runs before YOLO inference.

5.1 Element Reference

Element Description	
1	AUTO CALIBRATE — Computes 14 image metrics, builds candidates, scores with YOLO, populates recommended filters.
2	Add Filter — Select type: CLAHE, PCB Color Norm, Brightness/-Contrast, Gamma, Blurs, Denoise, Canny, Threshold, Sharpen, Unsharp Mask, Morph Close.
3	Pipeline List — Filters in execution order. Click to edit parameters.
4	Up/Down — Reorder selected filter. Order matters.
5	Node Index — Shows selected filter position.
6	Remove/Toggle — Delete or enable/disable selected filter.
7	Parameter Editor — Editable parameters appear here. Updates preview in real time.
8	Grab Frame — Captures current camera frame as preview source.
9	DEPLOY PIPELINE — Sends pipeline to RUN and AOI tabs.

Element Description	
10	Save Pipeline — Saves to <code>pipeline.json</code> . Auto-saves on changes.
11	ORIGINAL Panel — Unfiltered source image. Zoom-synced with filtered panel.
12	FILTERED Panel — Pipeline output. Updates in real time.
13	Load Image — Load any image as preview source.
14	Preview Filters — Manually trigger pipeline render.
15	Fit Button — Resets zoom on both panels.

5.2 Workflow

1. **Get source image:** Click **Grab Frame** **8** (camera must be running) or **Load Image** **13**
2. **Auto-calibrate:** Click **AUTO CALIBRATE** **1** — system recommends filters automatically
3. **Review result:** Compare ORIGINAL **11** vs FILTERED **12**
4. **Fine-tune:** Click a filter in pipeline list **3**, adjust parameters in editor **7**
 - CLAHE: increase clip_limit for more contrast, adjust tile_size for local adaptation
 - PCB Color Norm: adjust green_desat (0.70 default) to suppress FR-4 substrate
 - Gamma: increase for dark images, decrease for overexposed
5. **Test changes:** Watch FILTERED panel update in real time
6. **Reorder:** Use Up/Down **4** arrows (order matters! denoise before CLAHE)
7. **Save:** Click **Save Pipeline** **10**
8. **Deploy:** Click **DEPLOY PIPELINE** **9** to activate in RUN tab

TIP

Pipeline tip: Start with AUTO CALIBRATE. If results aren't perfect, manually add CLAHE (for uneven lighting) or PCB Color Norm (for green boards). Denoise filters should come before contrast enhancement.

6 AOI Tab: Offline Batch Inspection

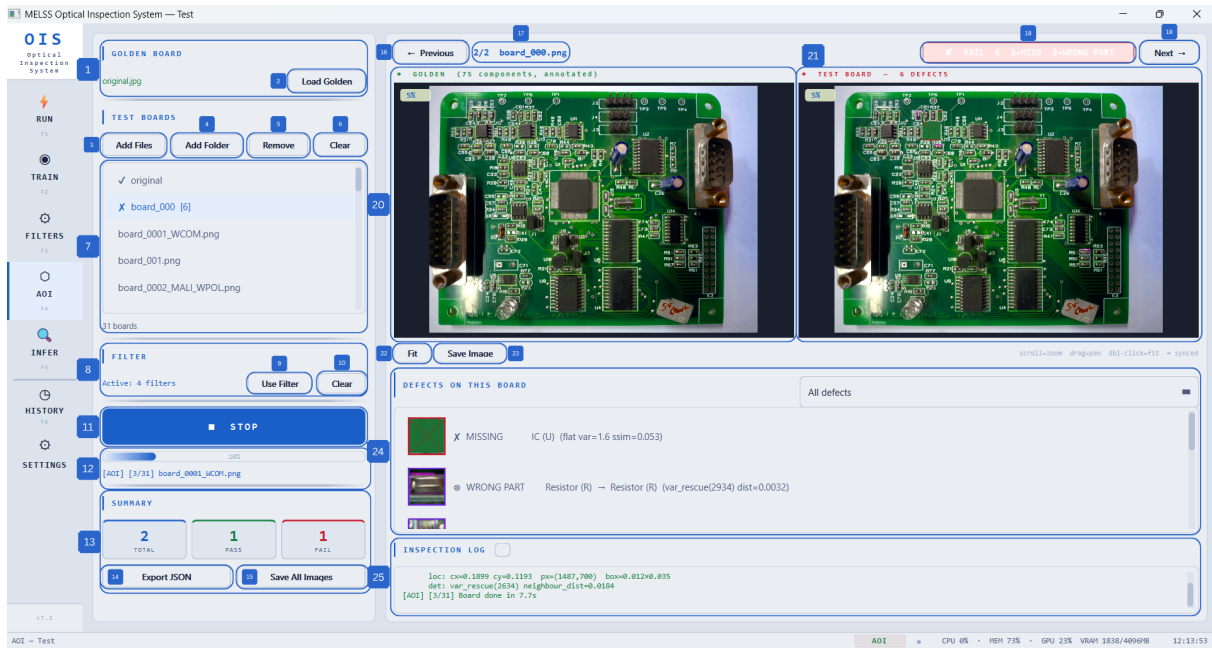


Figure 4: AOI Tab — Offline Batch Inspection

NOTE

Purpose: Run AOI engine on pre-captured board images against golden reference. No ROI editor included; operates on full image frame.

6.1 Element Reference

Element Description	
1	GOLDEN BOARD — Shows loaded golden reference. YOLO detects components to build reference list.
2	Load Golden — Select golden reference image.
3	TEST BOARDS — List of test board images. After inspection: ✓ for PASS, ✗ with defect count for FAIL.
4	Add Files — Multi-select file browser.
5	Add Folder — Add all images from a folder.
6	Remove — Remove selected board (Delete key works).
7	FILTER Section — Shows active filter pipeline status.
8	Use Filter — Apply pipeline from FILTERS tab.
9	Filter Status — Active filter count.

Element Description	
10	Clear Filter — Remove pipeline; use raw pixels.
11	RUN — Start batch inspection. Toggles to STOP.
12	Progress — Current board and overall progress.
13	SUMMARY — Total/PASS/FAIL cards. Updates real-time.
14	Export JSON — Save full report with per-board defects and signals.
15	Save All Images — Save annotated overlays for all boards.
20	Test Board List — Click any entry to view result.
16	Board Navigator — Previous/Next buttons.
17	Filename — Current board name.
18	Verdict Banner — PASS or FAIL with defect count.
19	Next — Advance to next board.
21	Dual Viewer — Left: Golden with green outlines. Right: Test with defect overlays. Zoom-synced.
22	Fit — Reset zoom on both panels.
23	Save Image — Save current result overlay as JPEG.
—	Defect List — Scrollable list with thumbnails, defect type, component label, signals. Dropdown filters by type.
—	Inspection Log — Live AOI engine decisions with full signal diagnostics.

6.2 Workflow

1. **Load golden:** Click **Load Golden** [2](#), select reference board image
2. **Add test boards:** Click **Add Files** [4](#) or **Add Folder** [5](#)
3. **(Optional) Apply filter:** Click **Use Filter** [8](#) to apply pipeline from FILTERS tab
4. **Run inspection:** Click **RUN** [11](#)
5. **Monitor progress:** Watch progress bar [12](#) and summary cards [13](#)
6. **Review results:** Click any board in test list [20](#) to see detailed results
 - Left panel = golden reference with green component outlines
 - Right panel = test board with defect overlays
 - Scroll to zoom both panels simultaneously
7. **Examine defects:** Scroll through defect list, click items to zoom to component
8. **Export:** Click **Export JSON** [14](#) for report, or **Save All Images** [15](#) for overlays

TIP

AOI tip: Use this tab to validate new filter pipelines or model versions before deploying to production RUN tab. Compare results against known PASS/FAIL boards.

7 INFER Tab: Single-Image Inference

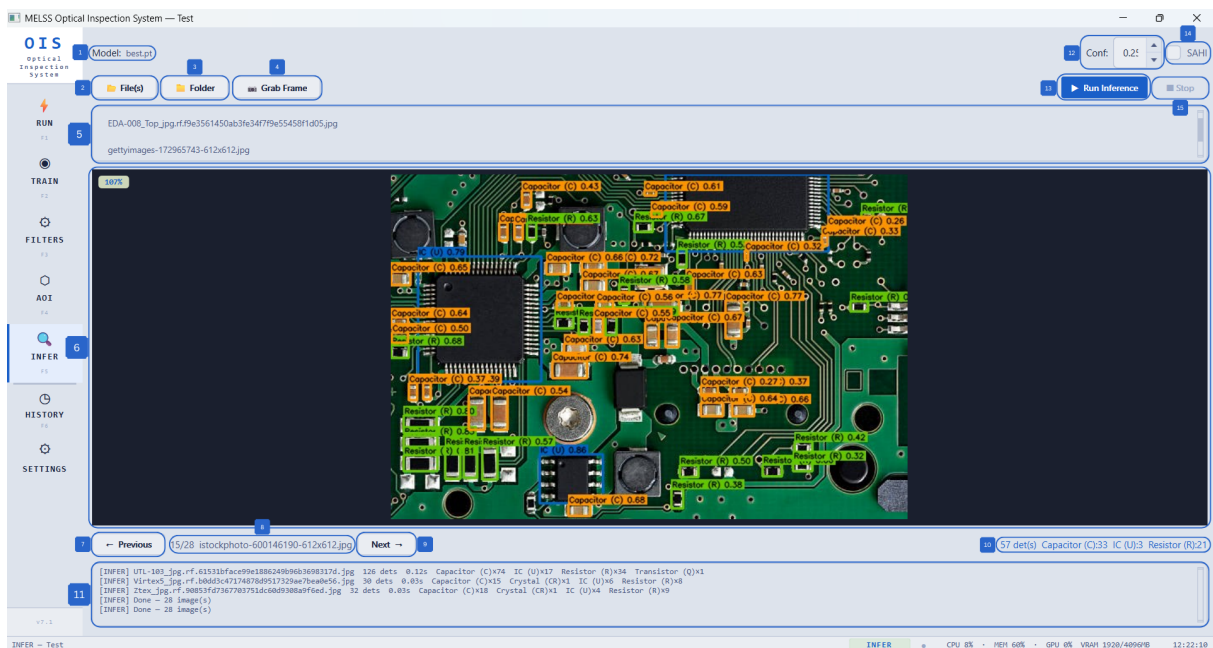


Figure 5: INFER Tab — Single-Image Inference

NOTE

Purpose: Run YOLO detection on individual images or folders. No golden comparison — pure object detection.

7.1 Element Reference

Element Description	
1	Model Path — Shows currently loaded model filename.
2	File(s) — Multi-select file browser.
3	Folder — Add all images from a folder.
4	Grab Frame — Capture current camera frame.
5	Queue List — Queued images. Click to view result.
6	Image Viewer — Detection overlays: colour-coded boxes with class and confidence.

Element Description	
7	Prev — Previous image.
8	Counter — Current position (N/Total).
9	Summary — Per-class detection count.
10	Log — Scrolling log of all processed images.
11	Confidence — Minimum confidence for detections (0.05-0.99, default 0.25).
12	Run — Process all queued images.
13	Stop — Cancel in-progress run.
14	SAHI — Enable for large images (>1920px). 3-5x slower but catches small components.

7.2 Workflow

1. **Add images:** Click **File(s)** [2](#), **Folder** [3](#), or **Grab Frame** [4](#)
2. **Set confidence:** Adjust **Confidence** spinner [11](#) (0.25 default)
3. **(Optional) Enable SAHI:** Check **SAHI** [14](#) for high-resolution images (>1920px)
4. **Run:** Click **Run** [12](#)
5. **Monitor:** Watch log [10](#) for progress
6. **View results:** Click images in queue list [5](#) to see detections
7. **Navigate:** Use Prev/Next [7](#) buttons or click directly in queue
8. **Stop if needed:** Click **Stop** [13](#) to cancel mid-run

TIP

INFER tip: Use this tab to verify a newly deployed model works correctly before running production inspection. The detection summary [9](#) shows per-class counts - useful for BOM verification.

8 HISTORY Tab: Inspection Records

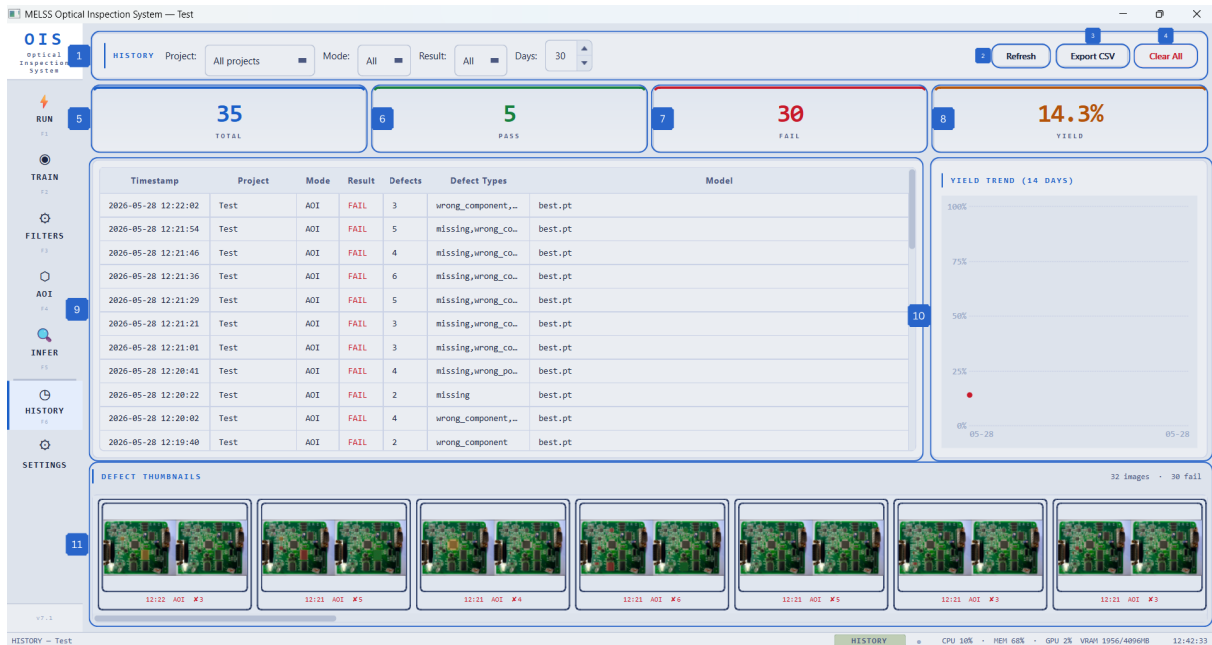


Figure 6: HISTORY Tab — Inspection Records

NOTE

Purpose: Review past inspections stored in SQLite database. Filter and export data.

8.1 Element Reference

Element Description	
1	HISTORY Header — Section label. Full-width filter bar below.
2	Refresh — Re-query database and update display.
3	Export CSV — Export filtered rows to CSV.
4	Clear All — Delete records matching current filter. Irreversible.
5	TOTAL — Sessions matching filter.
6	PASS — Sessions with zero defects.
7	FAIL — Sessions with at least one defect.
8	YIELD — Yield percentage ($\text{PASS} \div \text{TOTAL} \times 100\%$).
9	Results Table — Timestamp, Project, Mode, Result, Defects, Defect Types, Model. PASS in green, FAIL in red.
10	Trend Chart — 14-day sparkline. Each dot is one day's aggregate yield.

Element Description

- 11** **Thumbnails** — Horizontally scrollable strip of annotated FAIL board thumbnails. Click to open full image.
-

8.2 Filter Controls

Filter	Description
Project	Dropdown: All projects or specific project name
Mode	All / RUN (live) / AOI (offline)
Result	All / PASS / FAIL
Days	Number of past days to include (default: 30)

8.3 Workflow

1. **Set filters:** Choose Project, Mode, Result, and Days
2. **Refresh:** Click **Refresh** **2** to apply filters
3. **Review summary:** Check TOTAL/PASS/FAIL/YIELD cards **5-8**
4. **Scan table:** Browse results table **9** for specific inspections
5. **View thumbnails:** Scroll through thumbnail strip **11**, click to open full images
6. **Check trend:** Look at 14-day trend chart **10** for yield patterns
7. **Export:** Click **Export CSV** **3** for quality reporting
8. **Clean up (optional):** Click **Clear All** **4** to delete old records (irreversible!)

TIP

History tip: Export CSV monthly for long-term quality tracking. A sudden drop in the 14-day trend chart usually indicates a physical change (camera moved, lighting changed, or wrong reel loaded).

9 Signal Glossary

Token	Meaning
<code>cx=N cy=N</code>	Normalised centre coordinates (0.0 to 1.0)
<code>px=(X,Y)</code>	Pixel-space coordinates
<code>box=WxH</code>	Normalised bounding-box dimensions
<code>flat var=N</code>	Variance <80 → slot empty → MISSING
<code>ssim=N</code>	Structural Similarity Index (0.0-1.0)
<code>var_rescue(N)</code>	Component present but YOLO missed → WRONG COMPONENT
<code>neighbour_dist=N</code>	Distance to nearest same-label detection
<code>recov_ncc=N gain=N at N°</code>	Rotational NCC. Gain>0.15 at non-zero angle → MISALIGNED
<code>goh_delta=N</code>	GOH polarity delta → WRONG POLARITY
<code>cross_polar_rescue</code>	Cross-class detection, polarity confirms rotation → WRONG POLARITY
<code>cross_flat_override</code>	Cross-class YOLO hit but pixel variance empty → MISSING
<code>edge_rescue(N)</code>	Edge density confirms presence → WRONG COMPONENT
<code>at_slot_ssim(N)</code>	Detection at correct position with low SSIM → WRONG COMPONENT
<code>ssim_low(N) var(N) tmpl(N)</code>	All signals low → MISSING

10 Defect Overlay Colours

Colour	Defect	What is drawn
Red	MISSING	Red cross at golden slot position
Orange	MISALIGNED	Orange box at slot with rotation annotation
Magenta	WRONG COMPONENT	Pink box at wrong component's position, thin outline on expected slot
Cyan	EXTRA	Cyan box at unexpected component
Green	OK	Green box at matched detection

11 Workflow Guides

NOTE

This section provides step-by-step workflows for each tab. Follow these sequences to operate the system efficiently.

11.1 RUN Tab Workflow (Live Inspection)

1. **Start camera:** Click **START INSPECTION** [21](#). Camera initializes (3-5 seconds). Status bar shows "RUNNING".
2. **Set confidence threshold:** Adjust **AI CONFIDENCE** [20](#) slider.
 - **40-50%:** Recommended for production - balances detection vs false positives
 - **30-40%:** Use when components are small or lighting is poor
 - **50-60%:** Use when false positives are frequent
3. **Load golden reference:** Click **UPLOAD MASTER** [22](#). Three ways:
 - (a) **Take Latest Frame:** Capture current camera view (camera must be running)
 - (b) **Choose Picture:** Load golden board image from disk
 - (c) **Upload JSON:** Load existing golden_master.json fileAfter loading, golden component count appears in GOLDEN MASTER panel [26](#).
4. **Position first board:** Place board under camera. Use corner bracket overlays [4](#) as alignment guides. Position consistently - even 2-3mm shift can cause false defects.
5. **Inspect:** System automatically processes each frame at 8fps. Watch:
 - **Video panel:** Shows bounding boxes around detected components
 - **Annunciator banner:** Green "PASS" or Red "FAIL" with defect reason
 - **YIELD display:** Updates after each board
 - **LAST 10 strip:** Shows recent results (green=PASS, red=FAIL)
6. **Flag problematic boards:** When a board shows unexpected defects:
 - (a) Click **FLAG** [23](#) immediately
 - (b) Image saves to `project/golden_boards/flag_timestamp.jpg`
 - (c) Use flagged images later for model retraining
7. **Manual single inspection:** Click **INSPECT NOW** [25](#) to run AOI on current frame without affecting continuous inspection counters.
8. **End batch:** Click STOP (same as START button). Review session statistics:
 - TOTAL boards inspected
 - PASS count and percentage
 - FAIL count
9. **Reset for next batch:** Click **RESET** [16](#) to zero all counters. Does NOT delete history database records.

TIP

Production rhythm: START → present board → wait for result (1-2 seconds) → remove board → present next. For high volume, use FLAG only on failures; PASS boards need no action.

11.2 TRAIN Tab Workflow (Model Training)

1. **Collect training images:** Minimum 20 images, recommended 40-60.
 - Capture under varying lighting conditions (morning, afternoon, overcast)
 - Include slight positional variations
 - Include representative component appearances (different manufacturers, ages)

Click **Upload** [2](#) or drag files into thumbnail grid [4](#).
2. **Auto-label:** Click **Auto-Label All** [7](#). System runs YOLO on all unlabelled images. Progress bar shows completion. This takes 30 seconds to 2 minutes depending on image count.
3. **Review and correct labels:** Click each thumbnail [4](#) (green=labelled, red=unlabelled).
 - Select class from dropdown [18](#) (25 component types available)
 - Or use quick buttons [19](#): P (orange), R (red), C (yellow), IC (blue)
 - **Draw missing boxes:** Click-drag-release on component
 - **Delete wrong boxes:** Right-click or select + Delete key
 - **Move misplaced boxes:** Click to select, then drag
 - Click **Save & Next** [23](#) after each image

Goal: Eliminate all red-bordered thumbnails.
4. **Verify label quality:** Check that:
 - Boxes tightly enclose components (not too loose, not cutting into component)
 - Class labels are correct (Resistor vs Capacitor mislabels confuse the model)
 - No overlapping boxes on the same component
5. **Run augmentation (optional but recommended):**
 - Enable checkboxes [8](#): Brightness (always ON), Rotation 90/180/270, Flip Horizontal, Noise, Blur
 - Click **Run Augmentation** [9](#)
 - System generates 3-5x more training images
6. **Configure training:**
 - **Epochs** [12](#): 50 for quick test, 120 for production
 - **Batch** [13](#): 4 (4GB GPU), 8 (8GB+ GPU), 16 (24GB GPU)
 - **Base model** [11](#): Shows which .pt file will be used as starting point
 - (Optional) Click **Advanced** [14](#) to adjust learning rates, augmentation probabilities

7. **Start training:** Click **TRAIN** [15](#).

- Training runs in background - UI remains responsive
- Progress bar shows completion percentage
- Log shows epoch-by-epoch metrics: box loss, class loss, mAP50, precision, recall
- Typical training time: 10-30 minutes for 50 epochs on 40 images

8. **Wait for completion:** Training finishes when:

- All epochs complete, OR
- Early stopping triggers (no mAP improvement for 'patience' epochs)

Best weights saved to `project/models/golden_ft/weights/best.pt`

9. **Deploy trained model:** Click **DEPLOY** [17](#).

- Model copied to active path in `config.json`
- Status label shows deployed model filename
- Model now used by RUN, AOI, and INFER tabs

10. **Validate deployment:** Switch to INFER tab or RUN tab. Test on known boards to verify improvement.

TIP

Training cycle: Label 20 images → Train 50 epochs → Deploy → Flag failures → Add flagged images to training set → Retrain from previous best. This closed loop continuously improves performance.

11.3 FILTERS Tab Workflow (Pipeline Design)

1. **Capture reference image:** Click **Grab Frame from Camera** [8](#) (camera must be running in RUN tab). OR click **Load Image** [13](#) to use a saved board photo.

2. **Verify source appears:** ORIGINAL panel [11](#) shows the raw image. Check that:

- Board is well-centered
- Lighting is uniform (no dark corners or bright hotspots)
- Components are visible (not washed out or too dark)

3. **Run auto-calibration:** Click **AUTO CALIBRATE** [1](#).

- System analyzes 14 image quality metrics
- Tests 8 candidate pipeline architectures
- Populates pipeline list [3](#) with recommended filters
- Takes 10-20 seconds

4. **Review auto-calibration result:** Compare ORIGINAL [11](#) vs FILTERED [12](#).

- **Good result:** Components sharper, board background subdued, contrast improved
- **Poor result:** Image too dark/bright, edges lost, noise amplified

5. **Fine-tune individual filters:** Click a filter in pipeline list [3](#). Parameter editor [7](#) appears.

- **CLAHE:**
 - Clip Limit: 1.5-3.0 (higher = more contrast)
 - Tile Size: 4-12 (smaller = more local adaptation)
- **PCB Color Norm:**
 - Green Desat: 0.50-0.85 (higher = more green suppression)
 - Component Boost: 1.2-1.6 (higher = more component colour saturation)
- **Gamma:**
 - <1.0 = brighten image, >1.0 = darken image
 - 1.2-1.5 for underexposed, 0.7-0.9 for overexposed
- **Bilateral Denoise:**
 - d=5-9 (diameter), sigma=30-80 (strength)
 - Higher values smooth more but lose fine details

Watch FILTERED panel [12](#) update in real time as you adjust sliders.

6. **Reorder filters:** Select a filter, click Up/Down [4](#) arrows.
 - **Correct order:** Denoise → CLAHE → Sharpen
 - **Wrong order:** CLAHE → Denoise (amplifies noise then tries to remove it - poor result)
 - **PCB Color Norm:** Usually first in pipeline (before CLAHE)
7. **Test with multiple images:** Load different board images using **Load Image** [13](#). Verify pipeline works consistently across:
 - Different lighting conditions
 - Different board positions
 - Different component densities
8. **Save pipeline:** Click **Save Pipeline** [10](#). Config saves to `project/pipeline.json`.
9. **Deploy to production:** Click **DEPLOY PIPELINE** [9](#).
 - Pipeline sent to RUN tab and AOI tab
 - Status bar shows "Filters: N active"
 - Golden reference cache invalidated (will be re-filtered on next inspection)
10. **Validate in RUN tab:** Switch to RUN tab, start inspection. Verify detection quality improved.

TIP

Pipeline tuning strategy: Start with AUTO CALIBRATE. If board has strong green background, add PCB Color Norm first. If lighting uneven, add CLAHE. If image noisy, add Bilateral Denoise BEFORE CLAHE. If edges soft, add Unsharp Mask at the end.

11.4 AOI Tab Workflow (Offline Batch)

1. **Load golden reference:** Click **Load Golden** [2](#). Select the reference board image (same as used in RUN tab). System runs YOLO and shows component count in header.
2. **Add test boards:**
 - **Single files:** Click **Add Files** [4](#), multi-select images
 - **Folder:** Click **Add Folder** [5](#), select directory containing board images
 - Supported formats: .jpg, .png, .jpeg, .bmp

Test list [3](#) shows filenames.
3. **(Optional) Remove incorrect boards:** Select board in list, click **Remove** [6](#) or press Delete key.
4. **Apply filter pipeline (if used in production):** Click **Use Filter** [8](#).
 - Copies pipeline from FILTERS tab
 - Status shows "Active: N filters"
 - Pipeline applied to both golden and test images for apples-to-apples comparison

Skip this step if no filters used in production.
5. **Start batch inspection:** Click **RUN** [11](#). Button changes to STOP.
6. **Monitor progress:**
 - Progress bar [12](#) shows overall completion
 - Status label shows current board: "[AOI] [3/50] board_023.jpg"
 - Log shows per-board results as they complete
 - SUMMARY cards [13](#) update in real time
7. **Review failed boards immediately:** As each board completes:
 - Click its entry in test list [20](#)
 - Dual viewer [21](#) shows:
 - Left: Golden with green component outlines
 - Right: Test board with defect overlays
 - Defect list shows each defect with thumbnails
 - Click any defect to zoom to that component location
8. **Navigate between boards:** Use **Previous/Next** [16](#) or click entries in test list.
9. **Export results when complete:**

- **Export JSON** [14](#): Saves full report with per-board defects, signal values, pass/fail. Use for quality management systems.
 - **Save All Images** [15](#): Saves annotated overlay images for every board to chosen folder.
 - **Save Image** [23](#): Saves only current board's overlay.
10. **Stop mid-run if needed:** Click STOP (same as RUN button). Processed boards up to that point are kept.

TIP

Batch inspection use cases:

- Validate new model version before deploying to RUN tab
- Audit a day's production offline
- Test filter pipeline changes on historical data
- Generate inspection reports for quality certification

11.5 INFER Tab Workflow (Standalone Detection)

1. Add images to queue:

- **Files:** Click **File(s)** [2](#), select one or more images
- **Folder:** Click **Folder** [3](#), select directory
- **Live frame:** Click **Grab Frame** [4](#) (camera must be running in RUN tab)

Queue list [5](#) shows all queued images.

2. Set detection confidence: Adjust **Confidence** spinner [11](#).

- **0.20-0.30:** Use for small components or when you want to see all possible detections
- **0.30-0.50:** Standard range, balances detection vs false positives
- **0.50-0.70:** Use when verifying only high-confidence detections

3. Enable SAHI for large images (optional): Check **SAHI** [14](#) if:

- Image width or height > 1920 pixels
- Board has very small components (0402, 0201 size)
- You need maximum detection accuracy and can tolerate slower processing (3-5x slower)

4. Run inference: Click **Run** [12](#).

- Log [10](#) shows progress: "[INFER] Loading model..."
- Each image processed sequentially
- Log shows per-image results with detection counts and timing

5. Monitor during run:

- Queue list shows processing status

- Click **Stop** [13](#) to cancel mid-run
- Results for completed images remain available

6. View results:

- Click any image in queue list [5](#) to view
- Image viewer [6](#) shows bounding boxes with class names and confidence scores
- Summary label [9](#) shows per-class counts (e.g., "57 det(s) Capacitor(C):33 IC(U):3 Resistor(R):21")
- Use **Prev/Next** [7](#) to navigate

7. Analyze detection patterns:

- Check if all expected components are detected
- Look for consistent missed components (may need more training data)
- Look for false positives (may need confidence adjustment)

8. Export images (manual): Use screenshot or save image viewer content as needed.

TIP

INFER use cases:

- Quick model validation after training
- Component counting for BOM verification
- Debugging why certain components are missed in production
- Comparing two model versions on the same image set

11.6 HISTORY Tab Workflow (Record Review)

1. Set filters:

- **Project:** Select specific project or "All projects"
- **Mode:** RUN (live inspections) or AOI (offline batches) or All
- **Result:** PASS, FAIL, or All
- **Days:** Number of past days to include (default 30)

2. Refresh view: Click **Refresh** [2](#). Table and cards update with filtered data.

3. Review summary cards:

- **TOTAL:** Number of sessions matching filters
- **PASS:** Sessions with zero defects
- **FAIL:** Sessions with at least one defect
- **YIELD:** Pass percentage ($\text{PASS}/\text{TOTAL} \times 100\%$) with colour coding

4. Browse results table: Scroll through **Results Table** [9](#).

- Columns: Timestamp, Project, Mode, Result, Defects, Defect Types, Model

- PASS shown in green, FAIL in red
 - Defect Types column shows comma-separated list (missing, wrong_component, wrong_polarity, misaligned)
5. **Examine trend chart:** Look at **Yield Trend Chart** [10](#).
 - X-axis: Last 14 days
 - Y-axis: Yield percentage (0-100%)
 - Each dot = one day's aggregate yield
 - Sudden drops indicate problems: lighting change, camera shift, wrong reel loaded
 6. **View failure thumbnails:** Scroll through **DEFECT THUMBNAILS** strip [11](#).
 - Shows annotated images for FAIL sessions only
 - Each card shows: timestamp, mode (RUN/AOI), defect count
 - Click any thumbnail to open full-size image
 - Use to spot recurring defect patterns
 7. **Export data:** Click **Export CSV** [3](#).
 - Saves currently filtered rows to CSV file
 - Includes: timestamp, project, mode, result, defect count, defect types, model
 - Use for quality management system integration
 8. **Clean up old records (optional):** Click **Clear All** [4](#).
 - Deletes records matching current filters
 - Confirmation dialog appears - this is IRREVERSIBLE
 - Export before clearing if you need to keep data

TIP**Quality monitoring routine:**

- Daily: Check YIELD card - should be stable day-to-day
- Weekly: Review thumbnails for new defect patterns
- Monthly: Export CSV, archive, analyze long-term trends
- Investigate immediately if 14-day trend drops >10%

12 Quick Start: Complete Setup from Zero

NOTE

Follow this exact sequence to go from first launch to production-ready inspection. Estimated time: 2-4 hours for first setup.

1. **Launch application:** Run `python optical_inspection_system.py`
2. **Create project:** RUN tab → SWITCH PROJECT → New Project → Enter name (e.g., "Product_Line_A")
3. **Capture training images (20-40 photos):**
 - Place board under camera
 - RUN tab → START INSPECTION
 - Adjust position to fill frame
 - For each image: click FLAG → reposition slightly → repeat
 - Images save to `project/golden_boards/`
4. **Generate initial labels:**
 - TRAIN tab → Auto-Label All
 - Wait for completion (1-2 minutes)
5. **Review and correct labels (30-60 minutes):**
 - Click each thumbnail in grid
 - Green border = labelled, Red border = needs labels
 - For each image: check all boxes, add missing ones, delete wrong ones
 - Click Save & Next after each image
6. **Run augmentation:**
 - Check: Brightness, Rotation, Flip Horizontal, Noise, Blur
 - Click Run Augmentation
 - Wait for completion
7. **Train model:**
 - Set Epochs = 120 (production quality)
 - Set Batch = 4 (for 4GB GPU) or 8 (for 8GB+ GPU)
 - Click TRAIN
 - Wait for completion (20-40 minutes)
8. **Deploy trained model:**
 - Click DEPLOY
 - Verify status shows deployed model filename
9. **Setup filter pipeline:**

- FILTERS tab → Grab Frame from Camera
- Click AUTO CALIBRATE
- Review FILTERED panel - if good, click DEPLOY PIPELINE
- If not good, manually adjust parameters until satisfied

10. **Capture golden reference:**

- RUN tab → START INSPECTION
- Place known-good board under camera
- Click UPLOAD MASTER → Take Latest Frame
- Verify component count appears in GOLDEN MASTER panel

11. **Validate:**

- Place same known-good board
- Should show PASS (green banner)
- If FAIL, check: confidence setting, filter pipeline, golden reference

12. **Start production:**

- Present boards continuously
- Monitor YIELD display
- Use FLAG button for any board showing unexpected results

13. **Continuous improvement loop:**

- After each shift, review flagged images
- Add flagged images to golden_boards folder
- Re-train from previous best.pt (50 epochs)
- Deploy new model

TIP

Success metrics:

- First setup: Expect 85-90% yield after initial training
- After 2-3 improvement cycles: 95-98% yield
- False positive rate: <5% of inspections
- False negative rate: <1% of defective boards

WARNING

Critical: Always recapture golden reference if:

- Camera is moved or refocused
- Lighting setup changes (different ring light, ambient light changes)
- Filter pipeline is significantly altered
- Board revision changes (different component placement)

Using an outdated golden reference is the chief cause of false defects.